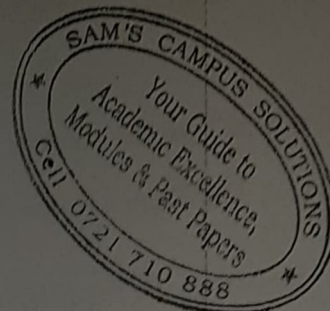
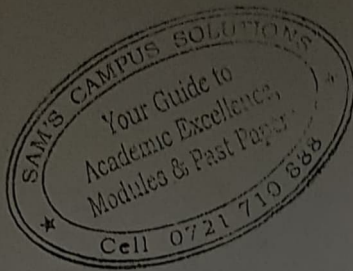




KENYA ITA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF

BACHELOR OF SCIENCE

ECU 105: ENGINEERING MATHEMATICS II

DATE: MONDAY 17TH DECEMBER 2018

TIME: 2.00 P.M. - 4.00 P.M.

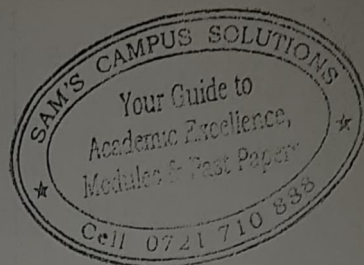
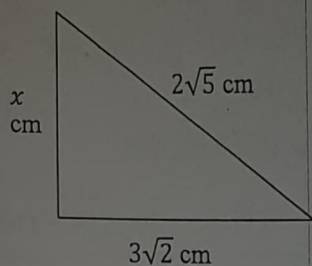
INSTRUCTIONS:

Question One (30 Marks)

- (a) Rationalize the denominator of $\frac{3-\sqrt{5}}{1+3\sqrt{5}}$ (5 marks)
- (b) Solve the equation $\log_3 x - 4 \log_x 3 + 3 = 0$. (3 marks)
- (c) Simplify $\frac{125^{\frac{2}{3}} \times 100^{-\frac{3}{2}}}{(\frac{5}{2})^{-2}}$ (3 marks)
- (d) A ball is thrown upwards from a roof top, 80m above the ground. It will reach a maximum vertical height and then fall back to the ground. The height of the ball from the ground at time t is h , given by $h = -16t^2 + 64t + 80$. What is the maximum height reached by the ball? (4 marks)
- (e) Solve the equation $\tan x = 2 \sin x$ for $0 \leq x \leq 360^\circ$. (3 marks)
- (f) In how many ways can 5 different toys be packed in 3 identical boxes such that no box is empty, if any of the boxes may hold all the toys? (3 marks)
- (g) (i) Expand $(1 - \frac{1}{2}x)^5$ upto the term with x^3 . (2 marks)
- (ii) Use the expansion to evaluate 0.998^5 to 4 decimal places. (3 marks)
- (h) Use De Moivre's theorem to show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$. (4 marks)

Question Two (20 Marks)

- (a) The diagram below shows a right-angled triangle.

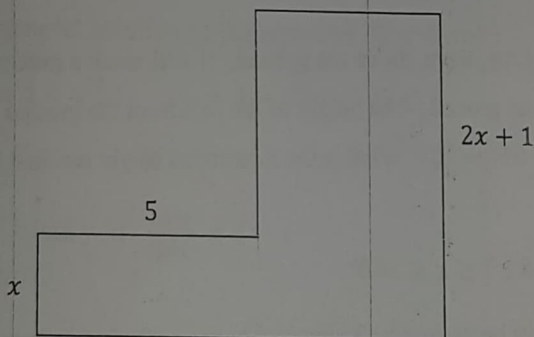


Find the value of x .

(3 marks)

- (b) The following picture shows the shape of a grass patch. If the area of the patch is $80m^2$, find x .

(5 marks)



- (c) In a certain country, the car number plate is formed by 4 digits from the digits 1, 2, 3, 4, 5, 6, 7, 8 and 9 followed by 3 letters from the alphabet. How many number plates can be formed if neither of the digits nor the letters are repeated?

(5 marks)

- (d) Evaluate $\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^{10}$

(7 marks)

ECU 105



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

ECU 105: ENGINEERING MATHEMATICS II

DATE: Thursday, 22nd February, 2018

TIME: 8.00 a.m. - 10.00 a.m.

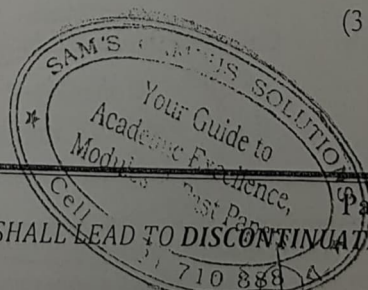
INSTRUCTIONS:

Answer question ONE and any other TWO.

QUESTION ONE [COMPULSORY - 30 MARKS]

- Solve for x : $8^{-\frac{x}{3}} \times 4^{x+1} \times 32^{-\frac{1}{5}} = 1$ (4 marks)
- The polynomial $ax^2 + bx + c = p(x)$ is divisible by x and $x + 1$. Find the values of a , b , and c . (3 marks)
- Write the complex number $z = \frac{2+i}{i}$ in polar form (3 marks)
- How many five digit even numbers greater than 6000 can be formed from the digits 1, 2, 3, 4, 5, 6, 7 and 8 with no repetition. (3 marks)
- If x is a real number find all possible values of x satisfying $\frac{2x-3}{x+4} < 1$. (4 marks)
- A committee of five is to be formed from four men and six women. Determine the number of ways this can be done if the number of women must exceed the number of men. (3 marks)
- Find the constant term in the expansion of $\left(2x - \frac{1}{4x}\right)^{10}$. (3 marks)
- Determine all solutions for the trigonometric equation $2\cos^2 x - \sin x - 1 = 0$ in the interval $[0, 2\pi)$. (4 marks)
- Solve for x
 $\log_6 3x + \log_6 (x-4) = 2$. (3 marks)

615.6



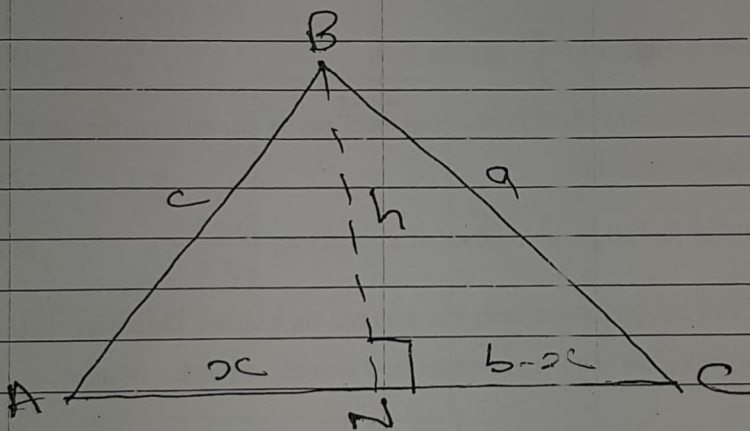
INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

QUESTION TWO

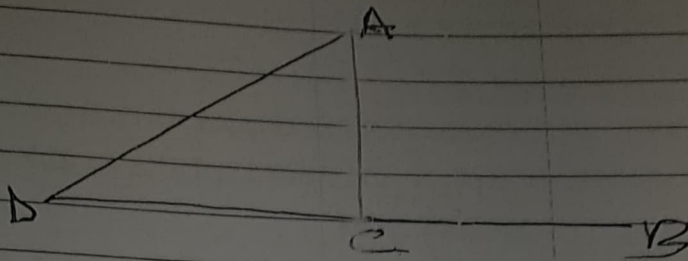
- a) Determine the set of values of x for which $12 - x - x^2 > 0$. (4 marks)
- b) The roots of the equation $2x^2 - 4x + 1 = 0$ are α and β . Find the equation with integral coefficients whose roots are $\alpha^2 - 2$ and $\beta^2 - 2$. (8 marks)
- c) Determine the line of symmetry, minimum or maximum value of x , x and y - intercepts of $f(x) = -2x^2 - 8x - 2$. Hence sketch the curve. (8 marks)

QUESTION THREE

- a) If $\sin x = \frac{4}{5}$ and $\cos y = \frac{12}{13}$ where x is obtuse and y is acute, find the values of $\tan(x + y)$. (4 marks)
- b) Factorize $\cos x - \cos 3x - \cos 5x + \cos 7x$. (6 marks)
- c) Consider the triangle below where A is an acute angle BN is perpendicular from B to AC and $AN = x$. Prove that $c^2 = a^2 + b^2 - 2ab \cos c$. (6 marks)



- d) Given that $DC = 9\text{cm}$, $AC = 7\text{cm}$, and $\cos \angle ACD = \frac{1}{9}$,
- Write down $\cos \angle ACD$. (2 marks)
 - Calculate the length of AD . (2 marks)

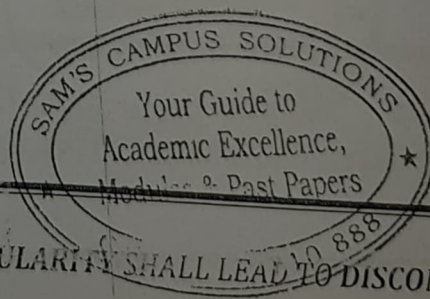


QUESTION FOUR

- a) How many six digit number can be formed using the digits 3, 4, 5, 6, 7 and 8 if
- each digit appears once in any number? (2 marks)
 - the numbers must be even and repetition of digits is not allowed? (3 marks)
 - the numbers are divisible by 5 and repetition of digits is allowed? (2 marks)
- b) i) Find the quotient and the remainder when $3x^4 - 5x^3 + x - 7 = P(x)$ is divided by $x - 3$. (4 marks)
- ii) Confirm your result in (i) above using the remainder theorem. (1 mark)
- c) Verify the identity
- $$\frac{\sin(x+y)}{\cos(x+y)} = \frac{\cot x + \cot y}{1 + \cot x \cdot \cot y} \quad (4 \text{ marks})$$
- d) A student wants to invite 12 friends but there is only room for 7. Find how many ways that she can choose whom to invite if two of them are twins and must not be separated. (4 marks)

QUESTION FIVE

- a) Find all the values of $z^5 + 32 = 0$ for the complex number z . (6 marks)
- b) Use Demorve's Theorem for complex numbers to prove that
- $$\cot 4\theta = \frac{1 - 6 \cot^2 \theta + \cot^4 \theta}{4 \cot^3 \theta - 4 \cot \theta} \quad (7 \text{ marks})$$



- c) Show that $\text{Log}_8 x = \frac{1}{3} \log_2 x$. Hence without tables or calculators, solve the equation $\text{Log}_2 (3x+1) + \text{Log}_8 (x-1)^3 = 6$ for $x > 0$.

(7 marks)



ECU 105 15th Dec/2014

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2014/2015

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN ENGINEERING (CIVIL, MECHANICAL, ELECTRICAL AND
ELECTRONICS)

ECU 105: MATHEMATIC ENGINEERING II

DATE: Monday, 15th December 2014

TIME: 4.30 p.m. – 6.30 p.m.

INSTRUCTIONS: ANSWER QUESTION ONE AND ANY OTHER TWO

QUESTION ONE – (30 MARKS)

1.a) Determine all the possible values of x if $2^{2x} - 42(2^x) + 320 = 0$ (4 marks)

i) State and prove the remainder theorem. (4 marks)

b) ii) State the remainder of the algebraic division

$$\frac{4x^5 - x^3 + x^2 - 3x + 3}{2x - 1} \quad (3 \text{ marks})$$

c) If the quadratic equation $3x^2 - 5x + 1 = 0$ has the roots α and B , determine the values of

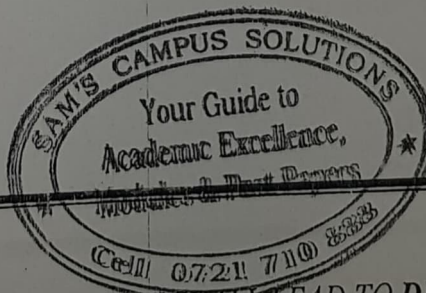
$$\left(\frac{\alpha^2}{B} + \frac{B^2}{\alpha} \right) + (\alpha - B)^2 \quad (6 \text{ marks})$$

d) i) Express $-5 + 5i$ into polar form. (3 marks)

ii) Hence or otherwise solve the equation $Z^2 + 5 - 5i = 0$ leaving your answer in polar form. (5 marks)

e) Verify the trigonometrical identity.

$$\frac{\sin \theta}{1 + \cos \theta} + \frac{\cos \theta}{\sin \theta} = \operatorname{cosec} \theta \quad (5 \text{ marks})$$



Page 1 of 3

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

QUESTION TWO – 20 MARKS

2.a) Simplify the following expressions

i) $3 \log_2 \left(\frac{5}{3} \right) - 2 \log_2 \left(\frac{10}{9} \right) + \log_2 \left(\frac{1}{30} \right)$

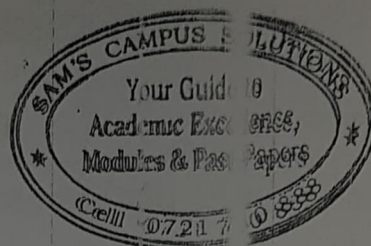
(3 marks)

ii) $\left(\frac{4}{9} \right)^{-\frac{1}{3}} + \left(12 \frac{1}{4} \right)^{-\frac{1}{2}}$

(3 marks)

iii) $\frac{\log 2 + \log 3}{\log \sqrt{216}}$

(3 marks)



b) Solve the equation

$(\log_3 x)^2 = \frac{\log_3 x^5 + 3}{2}$ are

(5 marks)

c) The roots of the equation $x^2 - px + 8 = 0$ are α and $\alpha + 2$. Determine the two possible values of P.

(6 marks)

QUESTION THREE – 20 MARKS

3.a) Given that $x = 2 \cos \theta$, show that $\frac{\sqrt{4-x^2}}{x} = \tan \theta$ (Take positive square roots) (4 marks)

b) Solve the trigonometric equations where $0 \leq \theta \leq 360$.

i) $2 \sin \theta = \sqrt{3} \tan \theta$

(4 marks)

ii) $2 \cos 2\theta = 3 - 2 \cos \theta$

(4 marks)

c) Let $\sin A = \frac{5}{13}$ and $\cos B = \frac{8}{17}$ where A is obtuse and B is acute. Determine with calculator the value of

i) $\cos(A+B)$

ii) $\tan(A-B)$

(8 marks)

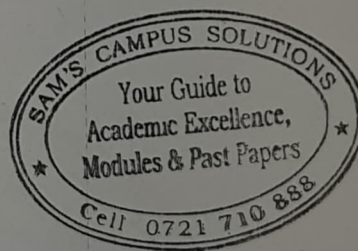
QUESTION FOUR – 20 MARK

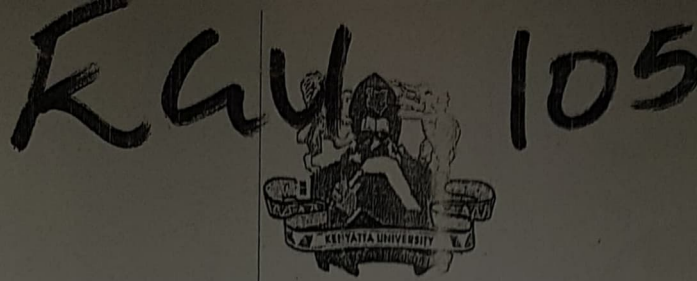
ECU 105 15th Dec 2014

- 4.a) Prove that $nc_r + nc_{r-1} = (n+1)C_r$ (5 marks)
- b) A mixed KU volleyball team containing 8 men and 6 women is to be chosen from 10 men and 9 women. In how many ways can this be done? (6 marks)
- c) Determine the number of permutations of all the letters in the word ENGINEERING. (3 marks)
- d) Explain $(1+x)^{\frac{1}{2}}$ upto 4 terms. Hence use the expansion to approximate find the value of $\sqrt{0.99}$. (6 marks)

QUESTION FIVE – 20 MARKS

- 5.a) Given that $Z_1 = 1-3i$, $Z_2 = 2+3i$ and $Z_3 = -1-i$. State the modulus and argument of:
- i) $Z_1 + Z_2$ (2 marks)
- ii) Z_3 / Z_2 (3 marks)
- iii) $(Z_1 - Z_3)^2$ (3 marks)
- b) Solve the equations below and show the roots in an argand diagram.
- i) $Z^3 = 8i$ (6 marks)
- ii) $Z^2 + 1 - \sqrt{8}i = 0$ and hence (6 marks)
- Factorise completely $Z^3 - 8i$ and $Z^2 + 1 - \sqrt{8}i$





KENYATTA UNIVERSITY
DEPARTMENT OF CIT

ECU 105: ENGINEERING MATHEMATICS II

CAT 1

TIME: 1HR

DATE: 28/10/2018

Answer ALL the Questions.

- 1) State the Remainder Theorem, hence find the remainder when $4x^3 - 5x + 1$ is divided by $2x - 1$. (3 marks)
- 2) State Factor Theorem hence solve the polynomial $x^4 - 3x^3 - 7x^2 + 15x + 18 = 0$ (6 marks)
- 3) Find the coefficient of the 6th term in the expansion of $(x+3)^{12}$ using binomial theorem (4 marks)
- 4) Expand $(1+3x)^{\frac{1}{2}}$ up to the fourth term (4 marks)
- 5) Simplify $\frac{1+i}{1-i} - (1+2i)(2+2i) + \frac{3-i}{1+i}$. (5 marks)
- 6) In how many ways can a group of 5 men and 2 women be made out of a total of 7 men and 3 women? (5 marks)
- 7) Find the number of words, with or without meaning, that can be formed with the letters of the word 'SWIMMING'? (3 marks)



Question Three (20 Marks)

(a) Evaluate:

(i) $\left(\frac{q^3 p^{\frac{-1}{2}}}{q^{\frac{-1}{3}} p} \right)^{\frac{3}{7}}$

(2 marks)

(ii) $\left(\frac{m^{\frac{1}{2}} n^{\frac{-1}{3}}}{n^{\frac{2}{3}} m^{\frac{-7}{4}}} \right)^{-\frac{1}{6}}$

(3 marks)

(b) Find the set of values for which $1 + 2x - 3x^2 < 0$.

(5 marks)

(c) When the expression $x^5 + 4x^2 + ax + b$ is divided by $x^2 - 1$, the remainder is $2x + 3$. Find the values of a and b .

(5 marks)

(d) Show that $\cot 2A = \frac{1}{2}(\cot A - \tan A)$.

(5 marks)

Question Four (20 Marks)

(a) Find the 8th term in the expansion of $\left(2x^2 - \frac{1}{x^2}\right)^{12}$

(4 marks)

(b) Find the general solution of the equation $\sin^2 4x = \frac{1}{4}$

(6 marks)

(c) Let $Z_1 = 1 + \sqrt{2}i$, $Z_2 = 3 - \sqrt{5}i$, find;

(i) $Z_1 Z_2$ (ii) $\frac{Z_1}{Z_2}$

(5 marks)

(d) Solve for x and y if $\frac{x+iy}{2+i} = 5 - i$

(5 marks)

Question Five (20 Marks)

(a) Solve for x in the equations $\log(8x) - \log(1 + \sqrt{x}) = 2$

(6 marks)

(b) A committee of 10 is to be chosen from 9 men and 6 women. In how many ways can this be done if at least 4 women are to be in the committee?

(5 marks)

(c) Expand $\frac{\sqrt{1+2x}}{1-x}$ upto the term including x^3 .

(6 marks)

(d) Obtain without using calculators the value of $\frac{1+\tan 60^\circ}{1-\tan 60^\circ}$

(3 marks)

- a) The roots of the quadratic equation $2x^2 - 7x + 4 = 0$ are α and β . Find an equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$. (5)

$$\text{Sum } (\alpha + \beta) = \frac{-b}{a} = \frac{7}{2}$$

$$\text{Product } (\alpha\beta) = \frac{c}{a} = 2$$

$$\text{New Product} \Rightarrow \frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 1$$

$$\text{New Sum} \Rightarrow \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{\left(\frac{7}{2}\right)^2 - 2(2)}{2} = \frac{33}{8}$$

$$x^2 - (\text{sum of roots}) + \text{product of roots} = 0$$

$$x^2 - \frac{33}{8}x + 1 = 0$$

$$8x^2 - 33x + 8 = 0$$

- b) The roots of a quadratic equation $x^2 + 3x - 2 = 0$ are α and β . Without solving the equation, find the quadratic equation whose roots are $\frac{1+\alpha}{\beta}$ and $\frac{1+\beta}{\alpha}$. (5)

$$x^2 - (-3)x + (-2)$$

$$\text{Sum } (\alpha + \beta) = -3$$

$$\text{Product } (\alpha\beta) = -2$$

$$\text{New Sum} = \frac{1+\alpha}{\beta} + \frac{1+\beta}{\alpha} = \frac{\alpha(1+\alpha) + \beta(1+\beta)}{\alpha\beta}$$

$$\frac{\alpha + \alpha^2 + \beta + \beta^2}{\alpha\beta} = \frac{\alpha + \beta + \alpha^2 + \beta^2}{\alpha\beta}$$

$$\frac{(\alpha + \beta) + (\alpha^2 + \beta^2) - 2\alpha\beta}{\alpha\beta} = \frac{-3 + (-3)^2 - 2(-2)}{-2} = -5$$

New product.

$$\frac{1+\alpha}{\beta} \times \frac{1+\beta}{\alpha} = \frac{(1+\alpha)(1+\beta)}{\alpha\beta} = \frac{1(1+\beta) + \alpha(1+\beta)}{\alpha\beta}$$

$$\frac{1 + (\beta + \alpha) + \alpha\beta}{\alpha\beta} = \frac{1 + (-3) + (-2)}{-2} = 2$$

$$x^2 - (-5)x + 2 = 0$$

$$x^2 + 5x + 2 = 0$$

- c) Solve the inequality $2x^2 + 4x \geq x^2 - x - 6$ (5)

$$2x^2 + 4x \geq x^2 - x - 6$$

$$2x^2 + 4x - x^2 + x + 6 \geq 0$$

$$x^2 + 5x + 6 \geq 0$$

$$x^2 + 3x + 2x + 6 \geq 0$$

$$x(x+3) + 2(x+3) \geq 0$$

$$(x+3)(x+2) \geq 0$$

$$\alpha = -3 \quad \beta = -2$$

	$x+3$	$x+2$	$(x+2)(x+3)$
$x \leq -3$	-ve	-ve	+ve
$-3 \leq x \leq -2$	+ve	-ve	-ve
$x \geq -2$	+ve	+ve	+ve

$\therefore$ The Set of Values for which $(x+2)(x+3) \geq 0$ are $(x \leq -3)$ and $(x \geq -2)$

Given that $\sin^{-1} x = \alpha$ and $\cos^{-1} y = \beta$, express $\tan(\alpha + \beta)$ in terms of x and y

$$x = \sin \alpha, y = \cos \beta$$

$$\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)}$$

$$= \frac{\sin \alpha \cos \beta + \sin \beta \cos \alpha}{\cos \alpha \cos \beta - \sin \alpha \sin \beta} = \frac{xy + \sin \beta \cos \alpha}{y \cos \alpha - x \sin \beta}$$

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\sin^2 \alpha = 1 - \cos^2 \alpha$$

$$\cos^2 \alpha = 1 - \sin^2 \alpha$$

$$\sin \alpha = \sqrt{1 - \cos^2 \alpha}$$

$$\cos \alpha = \sqrt{1 - \sin^2 \alpha}$$

$$\sin \beta = \sqrt{1 - \cos^2 \beta}$$

$$\cos \beta = \sqrt{1 - \sin^2 \beta}$$

$$\sin \beta = \sqrt{1 - y^2}$$

$$= \frac{xy + \sqrt{1 - y^2} \sqrt{1 - x^2}}{y(1 - x^2)^{\frac{1}{2}} - x(1 - y^2)^{\frac{1}{2}}}$$

Let $1 - y^2$ be a
Let $1 - x^2$ be b
And Rationalise

$$\frac{(xy + \sqrt{a} \sqrt{b})(y\sqrt{b} + x\sqrt{a})}{(y\sqrt{b} - x\sqrt{a})(y\sqrt{b} + x\sqrt{a})} \quad (a+b)(a-b) = a^2 - b^2$$

$$= \frac{xy\sqrt{b} + x\sqrt{a}\sqrt{b} + y^2\sqrt{b} + x^2\sqrt{a}}{y^2b - x^2a}$$

$$= \frac{y^2b - x^2a}{y^2(1 - x^2) - x(1 - y^2)}$$

Solve the following equations for $0 \leq x \leq 360^\circ$

i) $2 \sin 3x - 1 = 0$

$$\sin 3x = \frac{1}{2}$$

$$3x = 30^\circ \Rightarrow P_n \Rightarrow 0$$

$$180n + (-1)^n \times 30 = 30, 150, 210, 390, 510, 750, 870$$

$$x = 10, 50, 130, 170, 250, 290, 370$$

$$= 10^\circ, 50^\circ, 130^\circ, 170^\circ, 250^\circ, 290^\circ$$

Combination

$$\begin{aligned} &\rightarrow y^2 x \sqrt{b} + x a \sqrt{b} + y b \sqrt{a} + y x^2 \sqrt{a} \\ &\sqrt{b}(y^2 x + x a) + \sqrt{a}(y b + y x^2) \Rightarrow \text{numerator} \\ &\text{denominator} \rightarrow y^2(1 - x^2) - x(1 - y^2) \\ &y^2 - y^2 x^2 - x^2 + y^2 x^2 = y^2 - x^2 \\ &\text{Numerator} \rightarrow y^2 x \sqrt{1 - x^2} + x a \sqrt{1 - x^2} + y b \sqrt{1 - y^2} + y x^2 \sqrt{1 - y^2} \\ &(\sqrt{1 - x^2}(y^2 x + x a) + \sqrt{1 - y^2}(y b + y x^2)) \\ &= \frac{x \sqrt{1 - x^2} + y \sqrt{1 - y^2}}{y^2 - x^2} \end{aligned}$$

ii) $2 \sin^2 x + 3 \cos x - 3 = 0$

$$2(1 - \cos^2 x) + 3 \cos x - 3 = 0$$

Let $\cos x$ be y

$$\therefore 2 - 2y^2 + 3y - 3 = 0$$

$$-2y^2 + 3y - 1 = 0$$

$$2y^2 - 3y + 1 = 0$$

$$2y^2 - 2y - y + 1 = 0$$

$$2y(y - 1) - 1(y - 1) = 0$$

$$(y - 1)(2y - 1) = 0$$

$$y = 1 \text{ or } \frac{1}{2}$$

$$\cos x = 1$$

$$x = 0$$

$$360n \pm 0$$

$$360(0) \pm 0 = 0 \checkmark$$

$$360(1) \pm 0 = 360 \checkmark$$

$$\cos x = \frac{1}{2}$$

$$x = 60$$

$$360n \pm 60$$

$$= 60, 300$$

$$x = 0, 360, 60, 300$$

a) If $Z_1 = 1 - 3i$ and $Z_2 = -2 + 5i$, determine $\frac{Z_1 Z_2}{Z_1 + Z_2}$ in

$(x + ib)$ form.

$$Z_1 Z_2 = (1 - 3i)(-2 + 5i)$$

$$\frac{1}{Z}$$

$$(-2 + 5i)$$

(4)

c) Find the first four terms in the expansion of $(1 - 8x)^{1/2}$ in ascending powers of x . Substitute $x = \frac{1}{100}$ and find an

approximation for $\sqrt{23}$ correct to five significant figures

$$1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \frac{n(n-1)(n-2)(n-3)}{4!}x^4 \quad (6)$$

$$1 + \frac{1}{2}x$$

$$+ \frac{1}{2}x$$

$$+ \frac{1}{2}x$$

$$+ \frac{1}{2}x$$

$$+ \frac{1}{2}x$$

$$(2)$$

$$23$$

$$1 + 1$$



b) In how many ways can a class of 20 children be split into two groups of 8 and 12, if there are twins in the class who must not be separated?

(4)

$$20C8 \times 20C12$$

$$18560$$

Case 1

$$\frac{1}{8}$$

d) Determine the specified term in the expansion of

$$(5x - 2y)^8$$

i $x^2 y^6$ term

ii. Fourth term.

(6)

$$(5x - 2y)^8$$

f) Factorize the expression $6x^3 - 17x^2 - 4x + 3 = 0$ and hence solve for x . (5)

e) If a polynomial $p(x)$ divided by $(x + 1)$ the remainder $r(x)$ is 4 and when $p(x)$ is divided by $(x - 2)$ the remainder $r(x)$ is 3, find $q(x)$ and $r(x)$ when $p(x)$ is divided by $(x^2 - x - 2)$ (5)

x

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KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2020/2021

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
ELECTRICAL, MECHANICAL, CIVIL, PETROLEUM, AEROSPACE, BIOMEDICAL
& BIOSYSTEMS ENGINEERING

ECU 105: ENGINEERING MATHEMATICS II

DATE: THURSDAY 11TH MARCH 2021

TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS: Answer Question One and any other TWO Questions

QUESTION ONE (30MARKS)

- a) If $Z_1 = 1 - 3i$ and $Z_2 = -2 + 5i$, determine $\frac{Z_1 Z_2}{Z_1 + Z_2}$ in $(x + ib)$ form. (3 marks) ✓
- b) Find the set of values for which $2x^2 + 5x - 3 > 0$ (5 marks) ✓
- c) In how many ways can a class of 20 children be split into two groups of 8 and 12, if there are twins in the class who must not be separated? (3 marks) ✓ 225
- d) The roots of the quadratic equation $2x^2 - 7x + 4 = 0$ are α and β . Find an equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$. (3 marks) ✓
- e) Without using tables or calculator, find the value of $\sin 165^\circ$ (3 marks) ✓
- f) Factorize the expression $6x^3 - 17x^2 - 4x + 3 = 0$ and hence solve for x . (5 marks) ✓
- g) Solve for x and y given $3x + 2iy - ix + 5y = 7 + i5$ (3 marks) ✗
- h) i. Write down the simplified expansion of $(1 + x)^6$ (3 marks) ✓
ii. Use the expansion up to the fourth term to find the value of $(1.03)^6$ to the nearest one thousandth (2 marks) ✓

QUESTION TWO (20 MARKS)

- a) Prove that when a polynomial $f(u)$ is divided by $au + b$ where $a \neq 0$, the remainder is $f\left(-\frac{b}{a}\right)$. Hence find the remainder when $4u^3 - 6u + 5$ is divided by $2u - 1$. (5 marks)
- b) Sketch the graph of $g(x) = 3 \cos 2x$ and give its period and amplitude (5 marks)
- c) A personnel manager must select 5 clerks from a pool of 14 applicants. How many selections are possible? (3 marks)
- d) Three towns A, B and C are situated so that $AB = 60\text{km}$ and $AC = 100\text{km}$. The bearing of B from A is 060° and the bearing of C from A is 290° . Calculate:

- i) The distance BC
- ii) the bearing of B from C

(4 marks)
(3 marks)

QUESTION THREE (20 MARKS)

a) Evaluate

i. $\frac{6!2!}{8!}$

(2 marks)

- ii. How many even numbers greater than 2000 can be made from the integers 1,2,3,4 if each integer is used only once.

(3 marks)

b) Evaluate $\sin 75^\circ + \sin 15^\circ$ without using tables/calculator

(5 marks)

c) Use Pascal's triangle to expand and simplify the expression $(x - 2y)^7$

(4 marks)

d) If $\tan(A - B) = \frac{1}{7}$ and $\tan A = 3$, find the value of $\tan B$

(3 marks)

e) Express $e^{1-i\pi/4}$ in the form $a + ib$

(3 marks)

QUESTION FOUR (20 MARKS)

a) If $\tan x = \frac{3}{4}$, determine the value of $\sin 3x$

(5 marks)

b) Solve for x in the equation $\sin^2 4x = \frac{1}{4}$ for $0 \leq x \leq 360^\circ$

(7 marks)

c) Find all the values for z for which $z^5 = -32$

(6 marks)

d) Write down the sums and products of the roots of the equation $2x(x + 1) = x + 7$

Sum = -0.5
P = -3.5

(2 marks)

QUESTION FIVE (20 MARKS)

a) Express the following in terms of $\alpha + \beta$ and $\alpha\beta$;

i. $\alpha^3 + \beta^3$

(2 marks)

ii. $(\alpha - \beta)^2$

(3 marks)

b) A mixed hockey team containing 5 men and 6 women is to be chosen from 7 men and 9 women. In how many ways can this be done?

(5 marks)

c) Prove that $\frac{\cot \theta - \tan \theta}{\tan \theta + \cot \theta} = \cos 2\theta$

(5 marks)

d) Solve the equation $\log_5 x - 4\log_5(2x - 3) = 1$

(5 marks)

Handwritten diagram showing a right-angled triangle with sides labeled $\sin$, $\cos$, and $\tan$.